

Air Quality Dynamics in a Coastal Country: Spatiotemporal Analysis of Pollutants in Sri Lanka

H.Panchali.U Fonseka
Department of Earth Resource
Engineering
University of Moratuwa
Katubedda, Sri Lanka

and
Arthur C Clarke Institute for Modern
Technologies*
Katubedda, Sri Lanka
panchali.u.fonseka@gmail.com

Chaminda Samarasuriya
Department of Earth Resource
Engineering
University of Moratuwa
Katubedda, Sri Lanka
chaminda@uom.lk

Hongsheng Zhang
Department of Geography
The University of Hong Kong
Pokfulam Road, Hong Kong
zhanghs@hku.hk

Upaka Rathnayake
Department of Civil Engineering and
Construction
Atlantic Technological University
Sligo, Ireland
upaka.rathnayake@atu.ie

Abstract— This study examines the spatiotemporal dynamics of air pollutants, Sulfur Dioxide (SO₂), Carbon Monoxide (CO), Nitrogen Dioxide (NO₂), and the Absorbing Aerosol Index (AAI) in Sri Lanka from 2020 to 2024 using Sentinel-5P data. The socio-economic impact was assessed using NOAA-20 VIIRS data. Seasonal trend decomposition and Space-Time Cube analysis reveal distinct pollutant trends, with higher levels in coastal areas like Colombo due to industrialization and urban density, and inland variability driven by topography and localized emissions. Meteorological factors, especially rainfall play a key role in influencing these trends. However, the contribution of anthropogenic activities remains evident in shaping air quality patterns. The findings support sustainable urban planning and emission control strategies.

Keywords— Air pollutants, Spatiotemporal analysis, Coastal vs. inland regions

I. INTRODUCTION

Sri Lanka, an island nation in South Asia, is home to many coastal cities that serve as vital economic hubs while undergoing rapid urbanization and industrialization. Sri Lanka has experienced significant economic growth in recent decades, leading to increased industrial activity and urban sprawl, particularly in coastal areas [1]. This growth has exacerbated environmental challenges, including elevated concentrations of atmospheric pollutants. Due to urban pollution, cities such as Colombo, Kandy, and Kurunegala experience poor air quality [2]. In addition, polluted air masses drifting from India and other regions, carried by wind, also contribute to rising pollution levels[3].

Many countries including the USA, European Union countries, and China are addressing these challenges through emission reduction strategies, [2]. Up to date emission scenario needs to be identified for implementing such strategies. However, as a resource constrained country, Sri Lanka, faces limitations in ground-based data collection systems to understand the real status [2]. In such situations, satellite-based earth observation technologies provide a viable alternative, enabling effective monitoring of air pollutants.

Thus, this study utilizes satellite-derived data to analyze air quality across Sri Lanka.

Focusing on satellite data from 2020 to 2024, this research conducts a comprehensive trend analysis of key pollutants Nitrogen Dioxide (NO₂), Sulfur Dioxide (SO₂), Carbon Monoxide (CO), and aerosols. Of particular concern are SO₂, CO, and NO₂ [5], while the Absorbing Aerosol Index (AAI), which represents aerosol columnar loading, is critical due to contributions from sources such as desert dust, sea salt, black carbon, and organic carbon [4]. These pollutants not only degrade air quality and harm human health, causing reduced lung function and respiratory issues [2] but also impact meteorological patterns and climate.

While previous studies on air quality in Sri Lanka [2] have largely relied on ground-based measurements with limited spatial coverage, this study provides one of the first comprehensive spatiotemporal assessments using satellite-derived data. By utilizing multi-year Sentinel-5P observations, the analysis overcomes the limitations of sparse ground monitoring networks and enables broader insights into national and regional pollution trends. Specifically, the study aims to: (i) assess spatiotemporal changes in air pollution across Sri Lanka, (ii) identify air pollution hotspots, and (iii) evaluate the influence of socio-economic activities on emissions. The findings contribute to environmental knowledge, supporting agencies and researchers in promoting sustainable development, informed decision-making, and mitigating the negative effects of air pollution on public health and the environment in heavily polluted areas.

II. DATA AND METHODS

A. Air Pollutants and Emission Data

This study utilized gaseous pollutant data from satellite observations to investigate the spatial and temporal characteristics of air pollution over the Colombo District. The Table 1 summarizes the satellite products used in this study.

B. Study Region

Sri Lanka, in the Indian Ocean, is geographically located between latitudes 5°55'N and 9°51'N and longitudes 79°41'E

and 81°53'E as depicted in Fig. 1. Its unique location, tropical climate, and diverse topography make it an ideal study area for analysing environmental and socioeconomic dynamics. The country features a mix of coastal regions, urban centres, and mountainous areas, each contributing distinctively to the spatial variability of air quality..

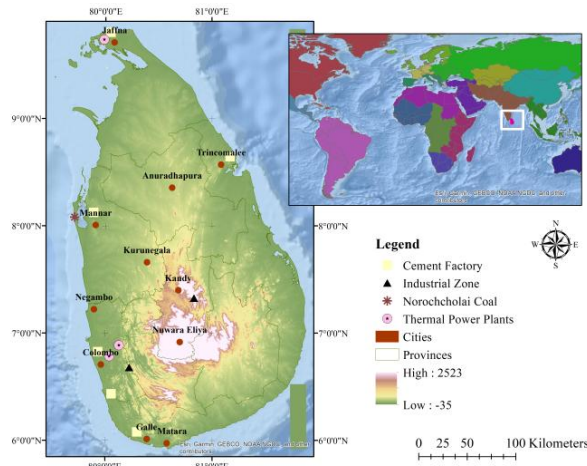


Fig. 1 Study Area

TABLE 1 SUMMARY OF DATA PRODUCTS

Data	Data Product	Spatiotemporal Resolution	Criteria of Data Screening	Data Source
SO ₂	Sentinel-5P NRTI SO ₂	7 km × 5.5 km; Daily	Cloud-free pixels; Values above detection thresholds	Copernicus Open Access Hub
NO ₂	Sentinel-5P NRTI CO	7 km × 5.5 km; Daily	Cloud-free pixels; Anomalously high or low concentrations	Copernicus Open Access Hub
CO	Sentinel-5P NRTI NO ₂	7 km × 3.5 km; Daily	Tropospheric columns; Cloud fraction below 0.3	Copernicus Open Access Hub
AAI	Sentinel-5P NRTI AAI	7 km × 3.5 km; Daily	Tropospheric columns; Cloud fraction below 0.3	Copernicus Open Access Hub
NTL	VIIRS Nighttime Lights (VNP46 A2)	500 m; Monthly, Annual	Filtered for background noise and stray light	NOAA National Centres for Environmental Information (NCEI)

C. Temporal Evaluation of Air Pollutants

The time-series data were analysed using a seasonal trend decomposition method [5], which broke down the data into trend, seasonal, and residual components. This approach enables a clearer interpretation of temporal variations by isolating long-term trends from recurring seasonal patterns and random fluctuations. It facilitates a better understanding of how pollutant concentrations evolve over time.

D. Spatial Evaluation of Air Pollutants

Mapping techniques implemented in Python were utilized to assess the spatial distribution of air pollutants. Monthly satellite-derived raster data were aggregated into annual datasets and then visualized to reveal spatial patterns and relationships that are often not apparent in the raw data. Spatial distribution maps highlighted the geographic variability in pollutant concentrations, emphasizing differences between coastal and inland regions.

E. Spatiotemporal analysis

A space-time cube analysis [6] was conducted using Python to integrate spatial and temporal data into a unified framework. This method enabled the detection of spatiotemporal trends and anomalies by examining changes in pollutant concentrations throughout the study period. The analysis facilitated the identification of consistent hotspots and emerging pollution patterns that may not be apparent in purely spatial or temporal evaluations.

F. Socioeconomic Evaluation Using NTL Data

NTL data were used as a proxy to analyse socioeconomic conditions across Sri Lanka. Pearson correlation analysis was performed to evaluate the relationship between NTL intensity and pollutant concentrations, helping to identify which air pollutants are most influenced by socioeconomic patterns such as urbanization and industrial activity.

III. RESULTS AND DISCUSSIONS

A. Temporal Evaluation of Air Pollutants

The monthly averages values were analyzed using seasonal trend decomposition method and summarized in Fig.2. All four pollutants exhibit fluctuations over time, with clear peaks likely corresponding to specific seasonal or temporal events.

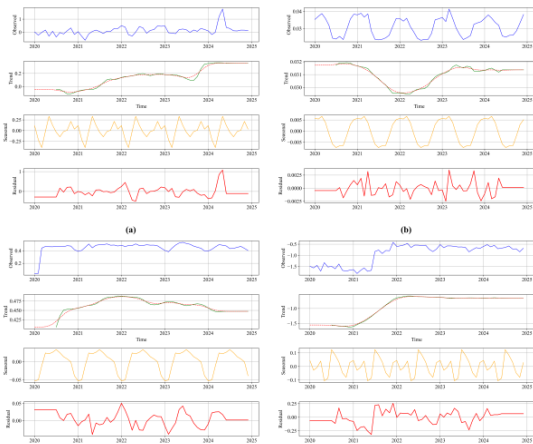


Fig. 2 Results of seasonal trend decomposition method (a) SO₂ (b) CO (c) NO₂ and (d) AAI

SO₂ as depicted in Fig.2a, declined until 2021 due to reduced industrial activity during the COVID-19 pandemic[7] but gradually increased post-2021 with the resumption of economic activities[7]. The CO trend, shown in the Fig.2b, exhibited a similar pattern, but followed a slightly different timeline. CO levels decreased until 2022, influenced by reduced vehicle usage due to the pandemic and the economic crisis[4,7], and rose thereafter as transportation and urban activities recovered.

In contrast, NO_2 as shown in Fig.2c displays a continuous decline over the timeline, reflecting long-term improvements in air quality. The differing trends of NO_2 , SO_2 , and CO can be attributed to variations in rainfall[8]. Reports indicate that Sri Lanka experienced increased rainfall during this period. Higher rainfall effectively removes NO_2 due to its greater solubility in water but has less impact on SO_2 and CO, which have lower water solubility[9]. CO, being a chemically stable gas with very low solubility in water, is minimally affected by rainfall in terms of atmospheric reduction [9]. As evidenced in the Fig.2d, AAI showed an increase after 2021, likely driven by construction, urban expansion, and biomass burning, before stabilizing at sustained levels. Strong seasonal variations across all pollutants highlight the influence of both natural and anthropogenic cycles. Additionally, anomalies in the data suggest localized pollution events caused by industrial surges, traffic congestion, or extreme weather events [4].

B. Spatial Evaluation of Air Pollutants

Fig. 3 illustrates the variation of air pollutants across both coastal and inland regions. As depicted in Fig.3a, higher concentrations of NO_2 are observed in Colombo, the primary economic hub of Sri Lanka, particularly near the Colombo Port[1]. This reflects the influence of maritime and urban emissions from dense vehicular traffic, industrial activities, and port operations [2]. In Fig.3b, CO follows a similar pattern, with higher concentrations along coastal regions, highlighting the contributions of urban and industrial activities in these areas. In contrast, the spatial and temporal distribution of SO_2 , as shown in Fig.3c, is more variable in inland cities, particularly in central and high-altitude regions. While inland hotspots do exist, coastal areas still exhibit relatively higher SO_2 concentrations due to industrial activities and maritime emissions.

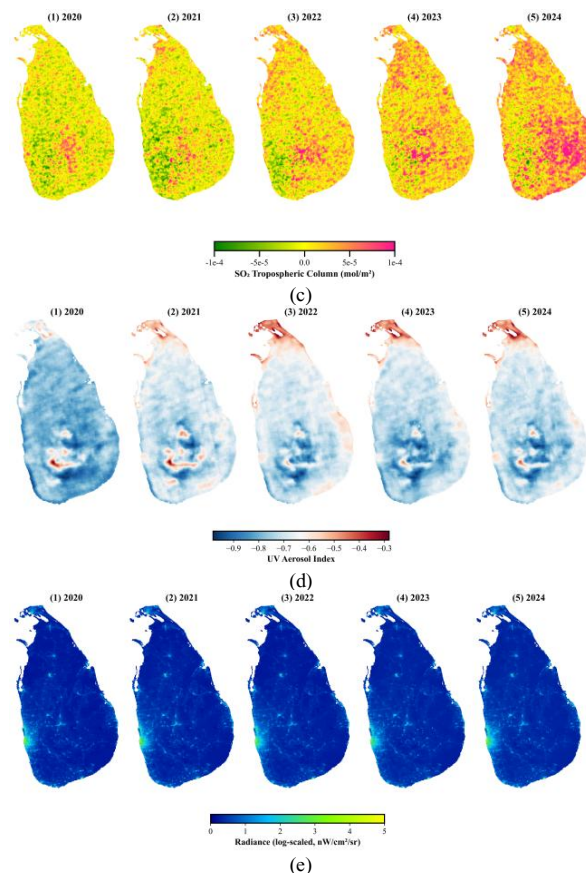
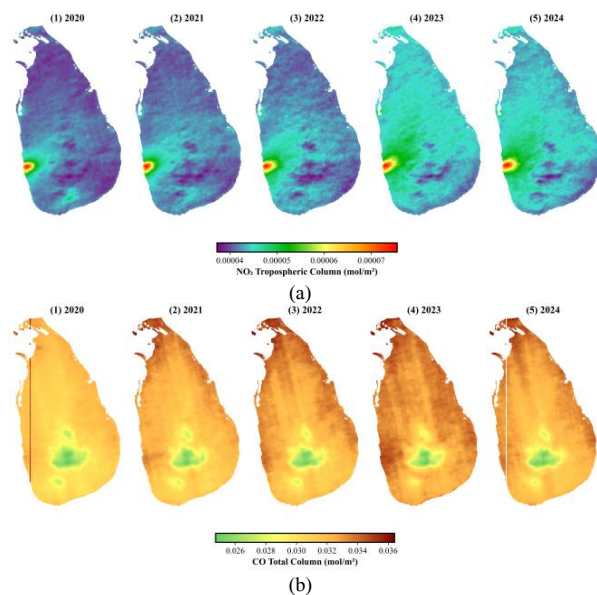


Fig. 3: Spatial Distribution of air pollutants (a) SO_2 (b) CO (c) NO_2 (d) AAI and (e) NTL

High AAI values are evident in both coastal and inland regions, as shown in Fig.3d, with more pronounced concentrations near coastal areas. The northern part of the country also exhibits elevated AAI values, likely due to proximity to South India and the Bay of Bengal. Winds carry dust from arid regions of India into northern Sri Lanka, increasing aerosol concentrations [3]. Fig.3e depicts socio-economic activities within the Colombo District. Brighter areas correspond to highly urbanized or economically active regions, primarily located in coastal areas, reflecting dense infrastructure, extensive transportation networks, and concentrated industrial zones[2].

C. Spatiotemporal evaluation

When analyzing the spatiotemporal patterns of air pollutants the country exhibit distinct vulnerabilities. NO_2 as depicted in Fig.4a concentrations peak in the Colombo region, indicating higher pollutant levels compared to other areas. Over time, these concentrations have increased, reflecting the rapid economic and industrial growth in this coastal hub, further exacerbating air quality issues[8]. CO follows a similar pattern as shown in Fig.4b, with noticeable increases in coastal regions such as Colombo. In earlier years, lower concentrations in central area suggest cleaner air and less urbanization, particularly in 2020.

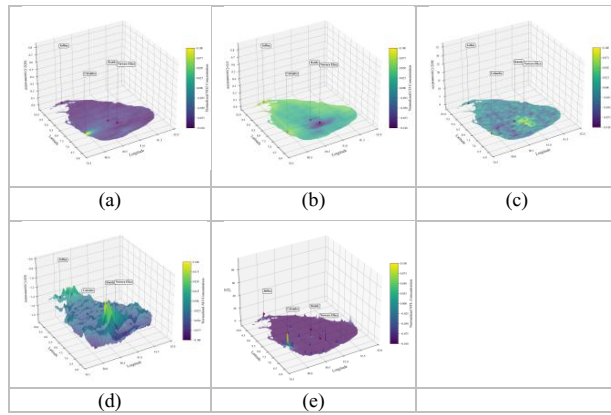


Fig. 4: Space-Time Cube Analysis from 2020 to 2024 (a) SO₂ (b) CO (c) NO₂ (d) AAI and (e) NTL

However, by 2024, CO levels have significantly increased. SO₂ concentrations remain relatively low across most areas, but isolated hotspots near coastal and industrial regions signal localized emissions of concern as visualized in the Fig.4c. While inland areas, such as valleys, may trap SO₂ due to topographic features [10], emissions in coastal cities can directly impact human health and the surrounding marine environment [10]. In Fig.4d, AAI shows notable increases, with significant contributions from urban and industrial emissions. Coastal cities like Colombo, with their high population densities and industrial zones[1], are hotspots for aerosol accumulation. Seasonal factors, such as sea breezes and thermal inversion, can exacerbate the trapping of aerosols in these coastal regions [2]. In contrast, inland mountainous areas like Kandy also exhibit high AAI values, but this is more likely due to topographic effects, such as thermal inversion and wind-driven dust [2,10]. Finally as depicted in Fig.4e data underscores the stark contrasts between urban and rural areas. Rural areas and inland regions exhibit low NTL intensity, reflecting limited infrastructure and sparse populations. This disparity highlights the socioeconomic divide, with coastal urban hubs bearing a disproportionate burden of air pollution due to their role as economic and industrial centers.

D. Correlation analysis with Socioeconomic activities

NTL data were used to reflect the socioeconomic conditions across cities in Sri Lanka, and its correlation with air pollutants was analysed. Table 2 presents the correlation values between NTL and various air pollutants over the years from 2020 to 2024. The socioeconomic influence on SO₂ fluctuates across the years. The negative correlation suggests that SO₂ is now less dependent on local sources and more influenced by transboundary pollution (e.g., from India) [3]. The declining trend further implies that external contributions are also decreasing. The correlation with CO remains relatively stable, except for peaks in 2020 and 2023, indicating that CO emissions may be linked to consistent sources, such as transportation. The correlation with NO₂ shows a generally steady pattern but declines slightly over the years with minor fluctuations.

TABLE 2 THE SUMMARY OF CORRELATION ANALYSIS

	2020	2021	2022	2023	2024
SO ₂	0.0317	-0.055	-0.0680	-0.0419	-0.0115
CO	0.112	0.0827	0.0851	0.111	0.0905
NO ₂	0.521	0.533	0.485	0.449	0.441
AAI	0.023	0.0288	0.0516	0.0214	0.0802

This stable relationship reflects the influence of transportation and industrial activities in urban areas[4], which are major sources of NO₂. The observed decline may be attributed to meteorological and atmospheric factors or effective urban pollution control measures. AAI fluctuates over the years, with notable peaks in 2022 and 2024. These peaks could indicate events like wildfires, industrial emissions, or increased urban activity that release black carbon or other absorbing aerosols [10].

IV. CONCLUSION

The spatiotemporal analysis of NO₂, CO, AAI, SO₂, and nighttime light (NTL) data reveals critical insights into the differences between coastal cities, inland areas, and Sri Lanka as a coastal country. These variations highlight the distinct environmental and socioeconomic challenges faced by coastal and inland regions in the context of urbanization, industrialization, and geographic features. Colombo, as a coastal city and the economic hub of Sri Lanka[1], consistently emerges as a hotspot for elevated pollutant concentrations and intense socioeconomic activity. Conversely, inland regions such as Kandy and central parts of the country display unique pollutant patterns[4]. As a coastal country, Sri Lanka faces unique challenges, as the majority of its urbanization and industrialization are concentrated in coastal areas. In this context, the data were validated using available ground data from the Central Environmental Authority. However, due to the reliance on sensor calibration and the absence of continuous data collection, the correlation was only moderate for some parameters. Addressing these challenges requires integrated approaches, including stricter emission regulations and targeted air quality management across the country. These efforts are essential for improving air quality, reducing health risks, and fostering balanced economic growth in both coastal and inland regions.

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